Mohammed Fayiz

ee25mtech11008@iith.ac.in | mhdfayiz122@gmail.com | +91 9946390185 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.41
N M H S S Thirunavaya	Class XII (BOARD OF HIGHER SECONDARY EXAMINATIONS)	2020	94.40%
T H S Kuttippuram	Class X (TECHNICAL EDUCATION DEPARTMENT)	2018	90.00%

WORK EXPERIENCE

MMRFIC TECHNOLOGY PVT LTD, Analog Design Intern, Bengaluru

- Working on Analog/Mixed-signal IC design using 130-nm technology.
- Currently designing a CMOS ring oscillator targeting a 120 MHz clock frequency.

SKILLS

Programming Languages : Verilog, Python

EDA Tools : Cadence Virtuoso, XILINX Vivado, LT-SPICE

RELEVANT COURSES

Analog Domain : Analog IC design , Analog IC design lab

Digital Domain : Digital IC Design, Digital IC Design Lab, CMOS Digital Design, Microprocessor and Computer Arch

Device Domain : Semiconductor Devices and Modelling, VLSI Technology

Memory Domain : Memory Circuits and System Design

PROJECTS

Low-Dropout (LDO) Voltage Regulator Design

Guided by Prof. Asudeb Dutta

- Designed a PMOS two-stage OTA-based LDO regulator in Cadence Virtuoso for output of 1.2V from supply of 1.8V
- Achieving 66 dB loop gain and 68.5° phase margin with load capacitance of 10p F for OTA.
- Performed line/load regulation, transient, PSRR, and PVT corner analysis.

Systematic Design of Two-Stage OTA Using gm/ID Methodology

Guided by : Prof. Asudeb Dutta

- Designed single-ended and fully differential two-stage OTAs using gm/ID-based transistor sizing and lookup-table methodology.
- Implemented and analyzed CMFB loops and Miller compensation, achieving approximately 53 dB Dc gain, 206 MHz UGB, Phase margin of 90°, Input common mode range of 0.56V to 0.84V.

1T-1C DRAM Array Design and Read/Write Simulation

- Designed and simulated a 1T-1C DRAM system including the storage cell, latch-type sense amplifier, 2-to-4 row decoder, and wordline level shifter.
- Verified complete write, precharge, sensing, restore, and refresh operations, demonstrating charge-sharing detection and restoration of the stored 1110 data pattern.

Dynamic StrongARM Latch Comparator Design

- Designed a dynamic StrongARM latch comparator for high-speed, low-power decision making, utilizing regenerative positive feedback for rapid amplification of small differential input signals.
- Performed transient and operating-region analysis to evaluate decision speed, output regeneration, and transistor behavior under clocked operation.

POSITIONS OF RESPONSIBILITY

Teaching Assistantship:

- TA for Digital circuits course Under **Prof. Asudeb Dutta** (July 2026-Present)
- TA for Electromagnetics course Under **Dr.Aneesh S** (Jan. 2026-May 2026)

EXTRACURRICULAR

Football, Volleyball, Swimming, Badminton